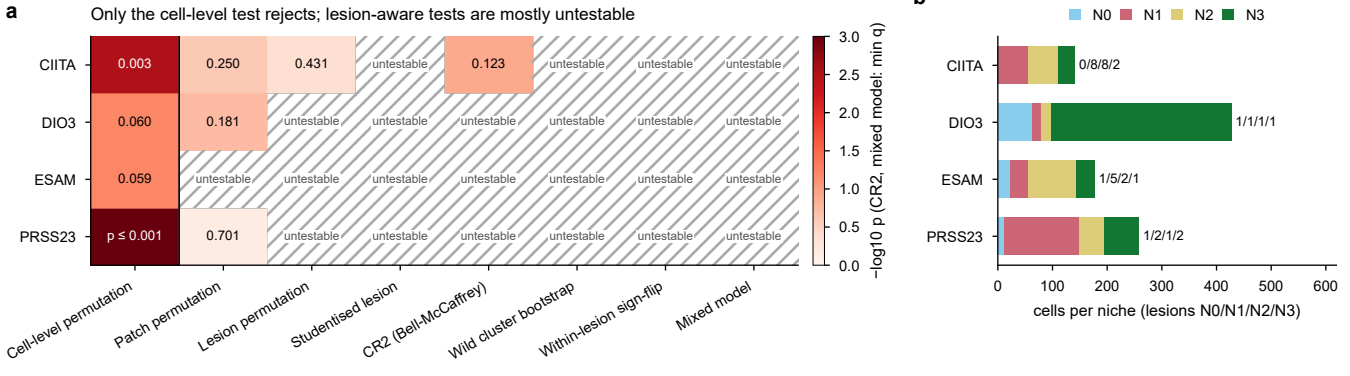
a
Only the cell-level test rejects; lesion-aware tests are mostly untestable
CIITA
DIO3
ESAM
PRSS23
0.003
0.250
0.431
untestable
0.123
0.060
0.181
0.059
p ≤ 0.001
0.701
Cell-level permutation
Patch permutation
Lesion permutation
Studentised lesion
CR2 (Bell-McCaffrey)
Wild cluster bootstrap
Within-lesion sign-flip
Mixed model
−log10 p (CR2, mixed model: min q)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
b
N0
N1
N2
N3
0/8/8/2
1/1/1/1
1/5/2/1
1/2/1/2
0
100
200
300
400
500
600
cells per niche (lesions N0/N1/N2/N3)